AI Usage Log (Annexure A format)

CE725 Team 8, Topic 2. Maintained as the work happens. Tool: Claude Code (Claude Fable 5.1) unless stated otherwise. Decision column: Accept / Modify / Reject, as decided by the team. Entries marked "team to confirm" were proposed by the AI and need a team member's sign-off.

No.	Date	Tool	Purpose / Prompt summary	Output summary	Decision	Reason and verification
1	2026-10-30	Claude Code	Read the course PDF, charter and readme and summarise what must be built	Correct list of deliverables and marking weights; noticed the charter PDF is truncated after "Intended users"	Accept	Checked against the PDFs; truncation confirmed by inspecting the 1-page file
2	2026-10-30	Claude Code	Asked whether the project instructions require an "agent"	Reported the word does not appear in any document; AI's role for Topic 2 is model structuring, formula checking and memo drafting	Accept	grep over all three documents returned no match
3	2026-10-30	Claude Code	Propose base case numbers, time step, platform and sensitivity drivers	Streamlit, monthly 36-month case, ₹20 cr land, 1 lakh sq ft, ₹3,000/sq ft cost, ₹7,500/sq ft price, 60% construction debt at 12%, 25% tax, 15% discount	Modify	Team accepted platform and case; debt definition later changed (entry 7)
4	2026-10-30	Claude Code	Write cash flow engine (land, approvals, S-curve construction, sales, booking-plus-instalment collections, debt sweep, tax)	feasibility/model.py; smoothstep S-curve $3x^2-2x^3$ chosen so it is reproducible in Excel	Accept (team to confirm)	Identity tests: spend totals, collections = revenue, loan fully repaid, equity = project + debt flows
5	2026-10-30	Claude Code	Write NPV, IRR, payback, profit on cost and break-even solver	feasibility/metrics.py	Modify	IRR was WRONG on first run: returned None for the base case. See entry 6
6	2026-10-30	Claude Code	Diagnose IRR = None	AI explained its own bisection bracket (−99% to +1000% per month) had the same NPV sign at both ends because of the final negative tax flow	Accept	Replaced with grid scan plus bisection; test asserts IRR = discount rate at the break-even price; Excel IRR() agrees to 1e−12
7	2026-10-30	Claude Code	Review base-case output	Peak debt only ₹0.79 cr and interest ₹0.04 cr because debt funded construction only and collections repaid it immediately	Modify	Economically unrealistic for a developer who borrows against land; changed loan to fund 50% of land, approvals and construction. Peak debt ₹12.1 cr, interest ₹2.0 cr
8	2026-10-30	Claude Code	Write sensitivity, tornado and scenario module	feasibility/sensitivity.py with 8 drivers, ±10%/±3 months/±2 pp swings, 4 preset scenarios	Accept (team to confirm)	Spot-check: price −10% reduces NPV by roughly revenue × 10% discounted, as expected
9	2026-10-30	Claude Code	Write investment memo generator	Template memo (offline) plus optional Claude memo restricted to a fact sheet, with a check that flags any number not in the fact sheet	Accept (team to confirm)	Template uses only computed values; LLM path not yet exercised (no API key on the build machine)
10	2026-10-30	Claude Code	Build Streamlit app	app.py with KPI rows and six tabs	Modify	First run crashed the cash flow table (mixed integer and "Total" index); fixed. Deprecated width argument replaced
11	2026-10-30	Claude Code	Generate Excel benchmark with formulas only	benchmark/build_benchmark.py → base_case_benchmark.xlsx; SUMPRODUCT used for lagged collections so instalment months stays adjustable	Accept (team to confirm)	Recalculated in LibreOffice; all 17 metrics match Python within 1e−11 (docs/validation_results.md)
12	2026-10-30	Claude Code	Automate Python vs Excel comparison	benchmark/compare.py	Modify	First version wrote to /tmp which snap LibreOffice cannot access; folder moved into project

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13	2026-10-30	Claude Code	Draft the 3-5 page user manual as a shared Claude Doc (11 sections: purpose, install, screen, inputs, results, cash flow, sensitivity, memo, validation, assumptions, limitations)	Manual drafted with base-case figures taken from the running app	Modify (team to confirm)	AI first wrote that land cost was the third-widest tornado bar; a check of the model showed sales launch delay ranks third. Corrected in the doc. Team to proofread and export to PDF
14	2026-10-30	Claude Code	Write property tests over 200 random input sets and compare NPV/IRR with numpy_financial	tests/test_properties.py; 383 tests total	Modify	AI's own monotonicity assertion "NPV always falls when the discount rate rises" failed on 5 cases; the MODEL was right (loss-making and non-conventional cash flows). Test restricted, learning recorded
15	2026-10-30	Claude Code	Write edge-case tests (invalid inputs, 100% debt, zero sweep, all cash at booking, loss-making case, late land, long sales window)	tests/test_edge_cases.py	Accept	All identities hold; loss case has zero tax and negative NPV as expected
16	2026-10-30	Claude Code	Test revealed the sensitivity table had 15 columns	AI explained the cause (per-driver step labels) and rewrote the table with positional columns and a Step column	Accept	Visible in screenshot docs/screenshots/03_sensitivity.png
17	2026-10-30	Claude Code	Memo number-check flagged "16.7%" in the template memo	Cushion percentage added to the fact sheet	Accept	check_numbers now returns no warnings for the template memo
18	2026-10-30	Claude Code	Run three more input cases through copies of the Excel workbook (compare_cases.py)	Cash flows matched in all cases. Excel IRR() gave #N/A in the downside case; heavy-debt case gave IRR −13.8% (Python) vs 839% (Excel)	Modify	Excel formula given a 1% guess. Multiple-IRR problem confirmed by locating both roots; tool now detects non-unique IRR and warns. Documented in validation report discrepancy 3 and 4
19	2026-10-30	Claude Code	Independent decimal recomputation of months 1, 2, 8, 12, 20, 30, 31, 36 and NPV from the written assumptions (hand_check.py)	Matches the engine to 1.5e−14	Accept (team to confirm)	Worked example for month 8 reproduced in the validation report; the team's calculator check is still required
20	2026-10-30	Claude Code	Browser-level test of the running app with Playwright and Chrome, with screenshots	24 checks: KPIs, every tab, metric switch, memo, Claude-mode error handling, break-even price gives NPV 0 and IRR 15%, validation error, reset, loss case, horizon extension, recompute time 1.1 s	Modify	Two test failures were test timing (edit immediately after Reset); helper now verifies the value. Screenshot review found inverted tornado colours for cost drivers, fixed
21	2026-10-30	Claude Code	AI's first multiple-IRR detector flagged the base case	Second root found at −98% per year (meaningless). Root count restricted to −10% to +50% per month and an NPV-sign consistency check added	Modify	Base case, no-debt and downside cases: 1 root; heavy-debt case: 2 roots, flagged
22	2026-10-30	Claude Code	Draft validation report, reflection report and presentation plan	docs/VALIDATION_REPORT.md, REFLECTION_REPORT_DRAFT.md, PRESENTATION_AND_DEMO.md	Modify (team to rewrite)	Reflection individual sections left empty on purpose; team must write in own words and verify every claim

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23	2026-10-30	Claude Code	Restart the app server from a script	Command killed itself because the kill pattern matched its own command line; the UI test ran against no server	Reject	Replaced with a kill by port number. Logged as a tooling error
24	2026-10-30	Claude Code	Browser timing check: one input change took 27.7 s	AI traced it to the sensitivity loops calling appraise() 56 times, each running two break-even bisections (300 ms) it did not need	Accept	Break-even made optional inside loops; recompute now 1.7 s. Would have stalled the live demo
25	2026-11-05	Claude Code	Add the team's AIML API key as a second memo provider (OpenAI-compatible gateway) alongside Anthropic direct	AI added a provider selector, a model field, key loading from `.streamlit/secrets.toml` (gitignored), and the same fact-sheet-only system prompt for both providers	Accept (team to confirm)	Key verified by listing the gateway's models; key stored only in a gitignored file; nothing else in the model changed
26	2026-11-05	Claude Sonnet 5.5 via AIML API	Write the base-case investment memo from the fact sheet	533-word memo in 7 sections, 12 s. Recommends "Proceed, with caution"; ranks price, construction cost, launch delay, land cost as drivers; notes a 3-month construction delay raises NPV; says the fact sheet does not define the scenarios	Modify	Number check: 0 of about 60 numbers unmatched. Read in full by a person. Memo was right that scenario definitions were missing: fact sheet extended, memo regenerated. Delay observation is correct for this model (no delay penalty) and is kept as a caveat. Saved verbatim in `docs/llm_memo_claude-sonnet-5-5.md`
27	2026-11-05	Claude Opus 5.5 via AIML API	Same prompt, same fact sheet, larger model	557-word memo, 17.5 s, same recommendation and same driver ranking	Accept (as a cross-check)	0 unmatched numbers. Two models agreeing on the narrative does not verify the numbers; the Excel benchmark does. Saved verbatim in `docs/llm_memo_claude-opus-5-5.md`
28	2026-11-05	Claude Code	Browser test failed 3 checks after the memo change	AI first suspected its own change, then measured: the Streamlit "Running" indicator appeared 1.2 s after the keystroke on a loaded machine and the test read the old value. The app was correct	Accept	Test fixed to wait for the indicator to appear before waiting for it to disappear; 27/27 pass. Lesson: a failing test is evidence about the test as often as about the tool
29	2026-11-05	Claude Code with web search	Data collection: find published sources for all 21 base-case inputs for a Mumbai (MMR) project	19 sources found: Knight Frank H2 2025 (Mumbai avg ₹8,856/sq ft), JLL 2024 land ₹17 cr/acre MMR, MCHI approval-cost paper (23% to 32% of project cost in BMC limits), RBI Oct 2026 repo 5.50%, EY/NSE 2024 real estate cost of equity 16.4%, s.115BAA 25.17%, RERA s.13 10% cap, NBFC construction finance 11% to 14%	Modify	Four primary documents (Knight Frank, EY/NSE, MCHI, RBI resolution) were opened and read; the rest were seen only as search summaries and are marked so in `docs/DATA_SOURCES.md`. Two pages (99acres, Business Standard) blocked automated access. The AI's first RBI figure (5.25%) was stale: the rate was raised to 5.50% on 7 Oct 2026, caught by cross-checking rbi.org.in. CPWD PAR 2025 could not be retrieved; a blog's unverified figure was rejected
30	2026-11-05	Claude Code	Decide whether to change the base case to the sourced values (marketing 3% to 4%, instalments 18+ months, discount rate 16.4%)	AI recommended keeping the base case and documenting the gaps, because changing inputs would invalidate the Excel benchmark, validation report and screenshots two days before the viva	Accept (team to confirm)	Gaps are stated in DATA_SOURCES §4 and the manual §10. The sensitivity rows at +1 and +2 pp discount rate and +20% costs show the base case stays viable under the sourced values

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31	2026-11-05	Claude Code with web fetch	Open every source that had only been seen as a search summary and confirm the figure used	17 of 19 sources opened, including the RERA Act itself (s.13(1) and s.4(2)(l)(D) quoted verbatim). Five figures corrected: Panvel land asking range is ₹3.2 to 12 cr/acre (not 9.2); Mumbai standard construction cost ₹3,086 (not 3,150); the DNA brokerage article is from 2018 and the L&T schedule from 2014; the 99acres price ranges were replaced by four pages that could be opened, which add a *registered* Kalyan rate of ₹7,900/sq ft	Modify	Two sources still need a browser (Business Standard 2020; MDPI paper statistics). One search-summary claim was wrong: findcirclerate does not call its figures estimates. A first attempt to fetch the blocked pages with a browser user-agent was refused by the tooling as bot-check circumvention and not retried
32	2026-11-05	Dinesh Sai (manual check, no AI)	Open the two sources the AI could not reach	MDPI paper opened: section 4.7 gives median IRR hurdle 18% (SD 4.85%) and median MDC 20% (SD 1.41%). Business Standard 2020 article is paywalled	Modify	S14 confirmed and its row rewritten to the figures actually seen; the AI's quoted mean (18.17%) and range (10% to 30%) were dropped as unchecked. S11 withdrawn from the source list; the interest-rate row now rests on the RBI release and the Loan Bazaar rate card only
33	2026-11-05	Claude Code	First pass of the course's manual benchmark (months 1, 8, 36, NPV, break-even) ahead of the Validation Engineer's re-performance	Calculator-style recomputation with 4-decimal rounding, no engine calls for intermediates; months 1, 8, 36 reproduce exactly; NPV 7.4514 vs 7.4515 (rounding); break-even prices found by bisection through the workbook's own formulas (25 LibreOffice recalculations, 7 minutes) and written to Summary!B21/B22	Modify (team to confirm)	This does NOT satisfy the course requirement by itself: it was done by the AI. It gives Surya a line-by-line template and expected values. Any line Surya cannot reproduce by hand is a discrepancy for section 7
34	2026-11-05	Surya Theja (manual, no AI)	Re-perform the manual benchmark: months 1, 8 and 36, the NPV sum and the two Goal Seek prices, with a calculator and the workbook	All lines reproduced the first-pass values; Goal Seek gives the same break-even prices as the bisection in Summary!B21/B22	Accept	Course requirement met by a person. The AI's first pass (entry 33) served as the template; the AI was not used for any arithmetic in this step
35	2026-11-05	Claude Code	Verify the Streamlit Community Cloud deployment	First run of the browser test timed out: the cloud page wraps the app in an iframe, so the test now takes the app URL from an environment variable and targets the inner frame. All 27 checks then passed on the deployment, including the Claude memo through the key stored in the app's secrets	Accept	Verified against the live URL, not the local copy; recompute 3.6 s on the cloud instance
36	2026-11-05	Claude Code	Draft the five individual reflections (section 5 of the reflection report) from the project record	Five first-person drafts of about half a page each, grounded in the AI log, validation report, data source list and test report, with bracketed prompts where only the member can supply the detail	Modify (required)	Disclosed in the report itself. Each member must rewrite their reflection in their own words and sign it; an unedited AI draft submitted as a personal reflection would be the misconduct the course PDF describes

Entries still possible before submission: any formula the team disagrees with, feedback from a practising engineer or lender if obtained. 36 entries; the minimum is 20.